Connecting Galactic Properties and Massive Black Holes

Feras Natsheh, supervised by Elia Cenci & Mélanie Habouzit

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ABSTRACT

In this lab report, we investigate the coevolution of galaxies and their central supermassive black holes from redshift $z = 6$ to the present day ($z = 0$) using the IllustrisTNG50 and IllustrisTNG100 cosmological simulations. We characterize the star-forming main sequence (SFMS) by analyzing the specific star formation rate (sSFR) as a function of stellar mass for galaxies with $M_\star \geq 10^7\,\mathrm{M}_\odot$. The SFMS normalization declines by nearly two orders of magnitude from $z = 6$ to $z = 0$, while the scatter increases from 0.18 dex to 0.43 dex. We quantify the relationship between black hole mass (M_{BH}) and stellar mass across cosmic time, finding a tight correlation at all epochs with systematic evolution in slope (from $p \approx 0.26$ at $z = 6$ to $p \approx 0.79$ at $z = 0$) and small evolution in normalization (from $q \approx 6.56$ at $z = 6$ to $q \approx 7.57$ at $z = 0$). The black hole mass distribution evolves from a sharp peak at the simulation black hole seed mass ($\sim 10^6\,\mathrm{M}_\odot$) at high redshift to a broader distribution with a prominent high-mass tail by $z = 0$ due to black hole growth, primarily through accretion. Black hole accretion rates decline systematically with cosmic time and correlate strongly with star formation activity. By tracking individual galaxy histories across three mass bins (10^9, 10^{10}, and $10^{11}\,\mathrm{M}_\odot$), we demonstrate that massive galaxies experience rapid quenching events at $z \lesssim 1$, characterized by near-simultaneous drops in gas mass, star formation rate, and black hole accretion rate occurring within $\sim 160 - 320$ Myr. Temporal lag analysis reveals no measurable offset between these processes within the snapshots time resolution, indicating tight coupling. Our results provide a coherent picture of black hole-galaxy coevolution in which shared gas supply drives concurrent growth, and AGN feedback regulates massive galaxy evolution at late times, establishing the observed Black hole-galaxy scaling relations.

1 INTRODUCTION

Understanding how galaxies form and evolve over cosmic time is one of the central challenges in modern astrophysics. Observations reveal that galaxies exhibit tight scaling relations such as the Schmidt-Kennicutt law (Kennicutt 1998) (relating gas content to star formation rate) and the star-forming main sequence (Speagle et al. 2014) (SFMS, relating stellar mass to star formation rate). These empirical relationships suggest that fundamental physical processes regulate galaxy evolution across cosmic time. Observations have also revealed that supermassive black holes (SMBHs) with masses $M_{\mathrm{BH}} \gtrsim 10^6\,\mathrm{M}_\odot$ reside at the center of most galaxies (Kormendy & Ho 2013), and that the properties of these black holes correlate tightly with properties of their host galaxies, such as stellar mass and bulge velocity dispersion. These correlations suggest that black holes and galaxies do not evolve independently, but rather coevolve through processes that couple their growth. Understanding this coevolution is crucial because black holes, despite containing only a small fraction of their host galaxy's stellar mass, can profoundly influence galaxy evolution through energetic feedback from active galactic nuclei (AGN).

AGN feedback is the injection of energy and momentum into the surrounding gas from accreting black holes, and is thought to play a critical role in regulating star formation, particularly in massive galaxies. When gas accretes onto a central black hole, it releases tremendous amounts of energy that can heat, ionize, or expel gas from the galaxy, thereby suppressing further star formation and black hole growth. This feedback process may explain why the most massive galaxies in the local universe are predominantly quiescent (not forming stars), despite residing in dark matter halos with large reservoirs of gas.

Observational studies have established several key scaling relations between black hole mass and galaxy properties. However, observations are limited by selection effects, measurement uncertainties, and the difficulty of observing galaxy evolution directly over cosmic time. Cosmological hydrodynamical simulations offer a complementary approach, allowing us to follow the coupled evolution of dark matter, gas, stars, and black holes from high redshift to the present day under controlled physical assumptions. By comparing simulation predictions with observations, we can test our understanding of the physical processes governing galaxy and black hole growth.

In this study, we use the IllustrisTNG50 and IllustrisTNG100 simulations to investigate how galaxies and their central black holes coevolve across cosmic time. We address the following questions: (1) How does the star-forming main sequence (the relationship between stellar mass and star formation activity) evolve from $z = 6$ to $z = 0$? (2) How does the relationship between black hole mass and stellar mass evolve, and what does this reveal about the growth of the black hole population? (3) How tightly are black hole accretion and star formation coupled, and does this coupling change over time?

This report is organized as follows. Section 2 provides theoretical background on black hole growth and AGN feedback. Section 3 describes the IllustrisTNG simulations and the data used in our analysis. Section 4 presents our results on the star-forming main sequence, black hole scaling relations, and the temporal evolution of individual galaxies. Section 5 discusses our findings in the context of observations and theoretical models and outlines directions for future work.

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2 THEORETICAL BACKGROUND

Most of the information in this section is taken from the textbooks Cimatti et al. (2020) and Sparke & Gallagher (2010).

2.1 Galaxies and Star Formation

Galaxies are gravitationally bound systems composed primarily of stars, gas, and dark matter. Stars form from dense, molecular gas within galaxies through gravitational collapse. The availability of cold gas therefore sets the fuel supply for star formation. Galaxies with larger gas reservoirs can sustain higher star formation rates. Furthermore, more massive systems can host larger gas reservoirs due to their deeper gravitational potential wells. These physical dependencies give rise to the star-forming main sequence (SFMS), which is an emergent empirical relation betwen stellar mass and star formation rate.

2.2 Black Holes

Astrophysical black holes are regions of spacetime where gravity is so strong that nothing, not even light, can escape. They are characterized solely by their mass and spin. Supermassive black holes grow primarily through two channels: (1) accretion of surrounding gas, which is expected to form an accretion disk through viscous processes, and (2) mergers with other black holes. Over long timescales and for massive black holes, accretion dominates the mass growth. As gas falls toward a black hole, gravitational potential energy is converted to heat and radiation in the accretion disk surrounding the black hole. It is this luminous accretion disk that produces the observed electromagnetic radiation, not the black hole itself. The accretion rate ($\dot{M}$) quantifies the mass inflow rate through the disk, and the luminosity is related to this rate through $L = \epsilon_{\rm rad}\dot{M}c^2$, where $\epsilon_{\rm rad}$ is the radiative efficiency. This radiative efficiency controls both the feedback from the AGN and the effective growth rate of the black hole, since only a fraction $(1 - \epsilon_{\rm rad})$ of the infalling mass is actually added to the black hole.

2.2.1 *Black Hole-Galaxy Scaling Relations*

Edwin Salpeter and Yakov Zeldovich along with Igor Novikov independently suggested that AGN energy is powered by accretion onto a SMBH located at the center of the host galaxy, with matter (gas and stars) close enough to the SMBH to be accreted (Salpeter 1964; Zel'dovich 1964). The radiative energy is generated by the conversion of gravitational energy of the matter infalling with an accretion rate $\dot{M}$ and heated to high temperatures in an accretion disc surrounding the SMBH (Shakura & Sunyaev 1976). More observations of local galaxies since then have revealed tight correlations between black hole mass and various galaxy properties, most notably bulge mass and stellar velocity dispersion (Kormendy & Ho 2013). These scaling relations are typically expressed as power laws. For example, the $M_{\rm BH} - M_{\rm bulge}$ relation:

$$\frac{M_{BH}}{10^9 \mathrm{M}_\odot} = q\left(\frac{M_{\rm bulge}}{10^{11}\mathrm{M}_\odot}\right)^p \tag{1}$$

with slope $p \approx 1$ and normalization q that depends on the sample selection. These relations are remarkable because they connect objects differing in mass by at least three orders of magnitude and in size by eight orders of magnitude. The Schwarzschild radius (Schwarzschild 1916) of a 10^8 $\mathrm{M}_\odot$ black hole is $r_S \sim 10^{-5}$ pc, while the bulge scale is $\sim$ kpc.

The redshift evolution of these scaling relations raises questions: Do the relations exist at high redshift, or do they emerge gradually as the universe evolves? How does the scatter change with time? Our analysis addresses these questions using simulation data spanning $z = 6$ to $z = 0$.

2.2.2 *The Eddington Limit*

The Eddington limit (Eddington 1921) applies to any accreting object where radiation pressure can balance gravitational force. For a black hole of mass $M_{\rm BH}$ accreting spherically symmetric gas, the maximum sustainable luminosity is the Eddington luminosity:

$$L_{\rm Edd} = \frac{4\pi c G m_p M_{\rm BH}}{\sigma_T} \simeq 1.26 \cdot 10^{46}\left(\frac{M_{\rm BH}}{10^8 \mathrm{M}_\odot}\right) \mathrm{ergs}^{-1} \tag{2}$$

where c is the speed of light, G is the gravitational constant, m_p is the proton mass, and σ_T is the Thomson scattering cross section. At this luminosity, the outward force from radiation pressure on electrons equals the inward gravitational force, preventing further accretion. The corresponding Eddington accretion rate is:

$$\dot{M}_{\rm Edd} = \frac{L_{\rm Edd}}{\epsilon_{\rm rad} c^2} \tag{3}$$

where $\epsilon \approx 0.1$ is the radiative efficiency of the accretion process (fraction of rest-energy released as EM radiation by a gas parcel to reach the central accreting body). The Eddington ratio $f_{\rm Edd} = \frac{L}{L_{\rm Edd}}$, characterizes how rapidly a black hole is accreting relative to its theoretical maximum, though there also exist black holes that accrete above this theoretical limit (super-Eddington accretion). Typical AGN have Eddington ratios in the range $10^{-2} \lesssim f_{\rm Edd} \lesssim 1$.

2.2.3 *Bondi Accretion*

In practice, black hole accretion is governed by the availability of cold gas in the galaxy's central region. The commonly used approximation is is the Bondi-Hoyle-Lyttleton accretion rate (Bondi 1952):

$$\dot{M}_{\rm Bondi} = \frac{4\pi\lambda_c (G M_{\rm BH})^2 \rho_\infty}{c_\infty^3} \tag{4}$$

where ρ_∞ and c_∞ are the density and sound speed of the gas far from the black hole, where the distribution is assumed uniform and isothermal, and λ_c is a dimensionless factor of order unity accounting for geometric effects. The steep $M_{\rm BH}^2$ scaling makes this prescription very sensitive to black hole mass. However, the Bondi model is idealized and may not accurately describe accretion in real galactic environments, which why some large-scale simulations employ alternative prescriptions.

2.3 AGN Feedback

AGN feedback refers to the processes by which energy and momentum from an accreting black hole affect the surrounding interstellar and circumgalactic medium. This feedback can take multiple forms including radiation pressure, winds, and jets, and can heat or expel gas, thereby suppressing star formation. In massive galaxies, AGN feedback is thought to be the primary mechanism responsible for quenching star formation and transforming galaxies from actively star-forming systems into quiescent ones. The energy released during black hole growth can couple to the surrounding gas through several channels, broadly classified as radiative and mechanical feedback:

Radiative feedback occurs when photons emitted by the accretion

disk heat, ionize, or exert pressure on surrounding gas. For an accreting black hole, the total energy released as radiation over its growth history is $E_{\rm BH} = \langle\epsilon_{\rm rad}\rangle M_{\rm BH}c^2$, where $\langle\epsilon_{\rm rad}\rangle$ is the time-averaged radiative efficiency. To assess whether this energy budget is sufficient to affect the host galaxy, we compare it to the binding energy of the galaxy's stellar component. For a galaxy bulge with stellar mass $M_\star$ and velocity dispersion σ, the binding energy is roughly $E_\star \sim M_\star\sigma^2$. The ratio of these energies is:

$$\frac{E_{\rm BH}}{E_\star} = \frac{\langle\epsilon_{\rm rad}\rangle M_{\rm BH}c^2}{M_\star\sigma^2} \simeq 100\left(\frac{\langle\epsilon_{\rm rad}\rangle}{0.1}\right)\left(\frac{\sigma}{300\,{\rm km\,s^{-1}}}\right)^{-2}\left(\frac{M_{\rm BH}/M_\star}{0.001}\right) \tag{5}$$

$E_{\rm BH}$ can be $\sim$ 100 times larger than $E_\star$, demonstrating that black holes can easily release more than enough energy to unbind gas from their host galaxies.

Mechanical feedback involves the direction injection of momentum and kinetic energy through winds, jets, or expanding bubbles of host gas. When gas accretes onto a black hole and radiates energy E, the emitted photons carry momentum $p = \frac{E}{c}$, which can be transferred to the surrounding gas.

AGN feedback operates in different modes depending on the accretion rate:

Quasar mode (high-accretion regime): When $f_{\rm Edd} \gtrsim 0.01$, the black hole accretes efficiently and produces high radiative luminosity. In this regime, both radiative and mechanical feedback are important. Radiation pressure and AGN-driven winds can heat and expel gas from the galaxy center, interrupting the fuel supply to both the black hole and star-forming regions. This mode is thought to operate in gas-rich galaxies during phases of rapid growth, and the AGN is believed to self-regulate, alternating between powerful outbursts and more quiescent periods (Novak et al. 2011).

Radio mode (low-accretion regime): When $f_{\rm Edd} \lesssim 0.01$, the black hole accretes at low rates, and radiative output is minimal. Mechanical feedback dominates, typically in the form of jets that inflate radio lobes and heat the surrounding gas. This mode is particularly effective in massive elliptical galaxies with hot gas atmospheres, where the mechanical luminosity of the jets can balance radiative cooling, preventing gas from cooling, condensing, and forming stars.

The transition between these modes is not sharp, and real AGN may exhibit hybrid behavior. Understanding which mode dominates in different environments and at different epochs is essential for predicting how AGN feedback shapes galaxy evolution.

2.4 The Star-Forming Main Sequence

The star-forming main sequence (SFMS) is an emergent empirical relation showing that most star-forming galaxies at any given epoch lie along a relatively tight relationship between stellar mass and star formation rate (SFR), defined as the rate at which a galaxy converts gas into stars (typically measured in $M_\odot$/yr). This empirical relation (Speagle et al. 2014) can be expressed as:

$$\log_{10}\psi = \alpha(z)\log_{10}M_\star + \beta(z) \tag{6}$$

where ψ is the SFR, and α and β are redshift-dependent parameters (α is the slope of the SFMS, while β is the main-sequence normalization). Both parameters are functions of time ($\alpha(z)$, $\beta(z)$). Alternatively, the SFMS is often expressed in terms of the specific star formation rate (sSFR $\equiv \frac{\rm SFR}{M_\star}$):

$$\log_{10}{\rm sSFR}(M_\star, z) = \left[\alpha(z) - 1\right]\,\log_{10}(M_\star) + \beta \tag{7}$$

The SFMS has been observationally well-established across the redshift range $0 \lesssim z \lesssim 6$ (Speagle et al. 2014; Rodighiero et al. 2011). Key observational trends include: (1) The normalization of the SFMS declines systematically from high to low redshift, reflecting the cosmic decline in available cold gas. (2) The slope α approaches unity at $z \sim 1-2$, meaning sSFR is nearly independent of stellar mass during peak epochs of cosmic star formation (Rodighiero et al. 2011). (3) The scatter around the SFMS is relatively constant at $\sigma \sim 0.3-0.4$ dex across most redshifts in observed samples, though it varies with both stellar mass and sample selection.

Galaxies can be classified based on their position relative to the SFMS:

Main-sequence galaxies lie on or near the SFMS and represent the majority of the star-forming population, by definition of SFMS. While many maintain relatively steady star formation, some low-mass galaxies oscillate around the SFMS due to burst star formation.

Starburst galaxies lie above the SFMS by at least 1-2 orders of magnitude. These systems are undergoing enhanced star formation, often triggered by galaxy mergers or interactions that funnel large amounts of gas into the galaxy center. Starbursts are short-lived because they rapidly consume their gas reservoirs, and, primarily, because intense stellar feedback disrupts further star formation.

Quiescent galaxies lie well below the SFMS. These galaxies have effectively ceased forming stars, because their gas has been either exhausted, expelled, or heated to temperatures where it cannot cool efficiently.

The majority of the star formation in the low-redshift Universe takes place in moderately massive galaxies ($10^{10} - 10^{11}\,M_\odot$), typically in high surface brightness disc galaxies. Roughly 15% of all star formation takes place in galaxies that show some sign of an active nucleus. About 20 per cent occurs in starburst galaxies (Brinchmann et al. 2004). The existence and evolution of the SFMS raises fundamental questions: What regulates star formation to maintain this tight relationship? Why do some galaxies remain on the main sequence while others transition to quiescence? What role do black holes play in driving galaxies off the main sequence?

2.5 Galaxy Mergers

Galaxy mergers are encounters between two or more galaxies that lead to the formation of a single remnant. In the hierarchical structure formation paradigm of ΛCDM cosmology, mergers are fundamental to galaxy assembly. The outcome of a galaxy encounter depends on the relative velocities, masses, and pericentric distance of the interacting systems. When galaxies pass at high relative speeds, the result is a "fly-by" that perturbs both systems but does not lead to coalescence. When relative speeds are low enough, gravitational forces bind the systems together, leading to a merger on timescales shorter than the Hubble time.

When galaxies merge, their central black holes are also expected to eventually merge. In a major merger, both black holes sink to the center of the remnant through dynamical friction. They eventually form a binary black hole system that hardens through gravitational interactions with stars and gas, ultimately coalescing via emission of gravitational waves. During this process, gas funneled into the galaxy center can trigger a luminous AGN phase, potentially producing a quasar. This merger-driven black hole growth is thought to be particularly important at high redshift, when merger rates were higher. The connection between mergers, starbursts, and AGN activity is central to many galaxy evolution models. Mergers can simultaneously enhance star formation and black hole accretion by driving gas inflows, potentially explaining the correlation between star formation and black hole accretion rates.

3 COSMOLOGICAL SIMULATIONS

3.1 Overview of Cosmological Simulations

Beyond cosmological simulations that model representative volumes of the universe, there exist zoom-in simulations that focus on individual galaxies or small samples at much higher resolution. The advantage of cosmological simulations lies in their large volumes, which enable statistical studies of galaxy evolution across a wide range of masses, redshifts, and environments. Cosmological simulations model the formation and evolution of galaxies by solving the equations governing the coupled dynamics of dark matter, gas, stars, and black holes within an expanding universe. These simulations start from initial conditions consistent with observations of the early universe (e.g., from the cosmic microwave background) and evolve forward in time using numerical methods. The simulations must account for a wide range of physical processes, from the large-scale gravitational clustering of dark matter to small-scale processes like star formation and supernova feedback, which occur on scales far below the simulation's resolution. These unresolved processes are implemented using "sub-grid" models,which are parameterized prescriptions calibrated to match observed galaxy properties.

3.2 The IllustrisTNG Suite

IllustrisTNG (The Next Generation) is a suite of large-volume cosmological magneto-hydrodynamical simulations (Nelson et al. 2021) that builds upon and improves the original Illustris simulation (Genel et al. 2014). TNG includes updated treatments of several key physical processes, most notably black hole seeding, accretion, and feedback. The simulations assume a flat ΛCDM cosmology consistent with the Planck Collaboration et al. (2016): $\Omega_{\Lambda,0} = 0.6911$, $\Omega_{m,0} = 0.3089$, $\Omega_{b,0} = 0.0486$, $\sigma_8 = 0.8159$, $n_s = 0.9667$, $andh = 0.6774$. The TNG suite consists of three simulation volumes:

TNG50: A $(50\ \mathrm{Mpc})^3$ comoving volume with the highest mass resolution, enabling detailed study of galaxy structure and small-scale gas dynamics.

TNG100: A $(100\ \mathrm{Mpc})^3$ volume with intermediate resolution, designed to sample a representative volume of the universe while maintaining reasonable computational cost. TNG100 uses the same initial conditions as the original Illustris simulation, allowing for direct comparisons.

TNG300: A $(300\ \mathrm{Mpc})^3$ volume with lower resolution, optimized for studying large-scale structure and rare, massive objects.

All TNG simulations use the moving-mesh AREPO code (Springel 2010), which solves the equations of magnetohydrodynamics on an unstructured Voronoi mesh that moves with the gas flow. This approach combines advantages of both Lagrangian methods (adaptive resolution following the matter distribution) and Eulerian methods (accurate shock capturing and conservation of conserved quantities).

The simulations evolve from an initial redshift $z = 127$ to the present day ($z = 0$), with 100 snapshots saved at logarithmically spaced intervals in cosmic time. Each snapshot contains full information about the positions, velocities, masses, and other properties of all dark matter particles, gas cells, star particles, and black holes.

3.3 Physical Models in TNG

The TNG simulations include sub-grid models for the following processes relevant to our study:

Star formation: Gas above a density threshold and below a temperature threshold is eligible to form stars. Star formation is modeled stochastically following a Schmidt-Kennicutt-type relation, with the star formation rate depending on local gas density.

Stellar feedback: Massive stars inject energy, momentum, and metals into the surrounding gas through stellar winds and supernova explosions. This feedback heats gas, drives outflows, and regulates star formation, especially in low-mass galaxies.

Black hole seeding: Black holes are seeded with a fixed mass of $M_{\mathrm{seed}} = 10^6\,\mathrm{M}_\odot$ in dark matter halos. This prescription does not model the physical formation process (e.g., from Population III stellar remnants or direct collapse), but provides an initial black hole population that then grows through accretion and mergers.

Black hole accretion and growth: Black hole accretion rates are computed using a modified Bondi-Hoyle prescription that accounts for the multiphase structure of the interstellar medium. Accreting black holes grow in mass, and their accretion luminosity is computed assuming a radiative efficiency $\epsilon_{\mathrm{rad}} \approx 0.1$.

As for AGN feedback, TNG implements a two-mode feedback model:

Quasar mode (high accretion, $f_{\mathrm{Edd}} \gtrsim 0.01$): Energy is injected thermally into surrounding gas, heating it and driving outflows.

Kinetic wind mode (low accretion, $f_{\mathrm{Edd}} \lesssim 0.01$): Energy and momentum are injected as kinetic winds, which can drive large-scale gas outflows and maintain quiescence in massive galaxies.

Gas Cooling: Hot gas cools radiatively through atomic and molecular line emission, allowing it to condense and become available for star formation. Cooling rates depend on gas temperature, density, and metallicity.

The AGN feedback model in TNG has been calibrated to reproduce several key observational benchmarks, including the galaxy stellar mass function, the stellar-to-halo mass relation, and the black hole-galaxy scaling relations at $z = 0$ (Nelson et al. 2021).

Various astrophysical processes must also be considered to achieve a realistic galaxy population. Many of these are implemented through effective sub-resolution models. Figure 1 shows the various elements of cosmological simulations. Many of these are implemented through effective sub-resolution models. Most cosmological simulations do not include all of these physical models simultanelously.

3.4 Data Used in This Study

For this analysis, we use catalog data from IllustrisTNG50, IllustrisTNG100, and Illustris provided by Dr. Mélanie Habouzit. These catalogs contain pre-computed properties for all galaxies and black holes at each snapshot, including:

Galaxy properties: Stellar mass ($M_\star$), star formation rate (SFR), gas mass within the stellar half-mass radius (M_{gas}), and specific star formation rate (sSFR = SFR / $M_\star$).

Black hole properties: Black hole mass (M_{BH}), accretion rate ($\dot{M}$), and bolometric luminosity (L_{bol}).

Merger trees: These link galaxies across time, allowing us to track individual systems as they grow and evolve.

We focus on galaxies with stellar masses $M_\star \geq 10^7\,\mathrm{M}_\odot$ and analyze snapshots spanning redshifts $z = 0, 0.5, 1, 2, 3, 4, 5,$ and 6. The stellar half-mass radius ($r_{1/2}$) is defined as the radius enclosing 50% of the galaxy's stellar mass. Gas masses are measured within this radius to focus on the central, star-forming region where gas is available for both star formation and black hole accretion.

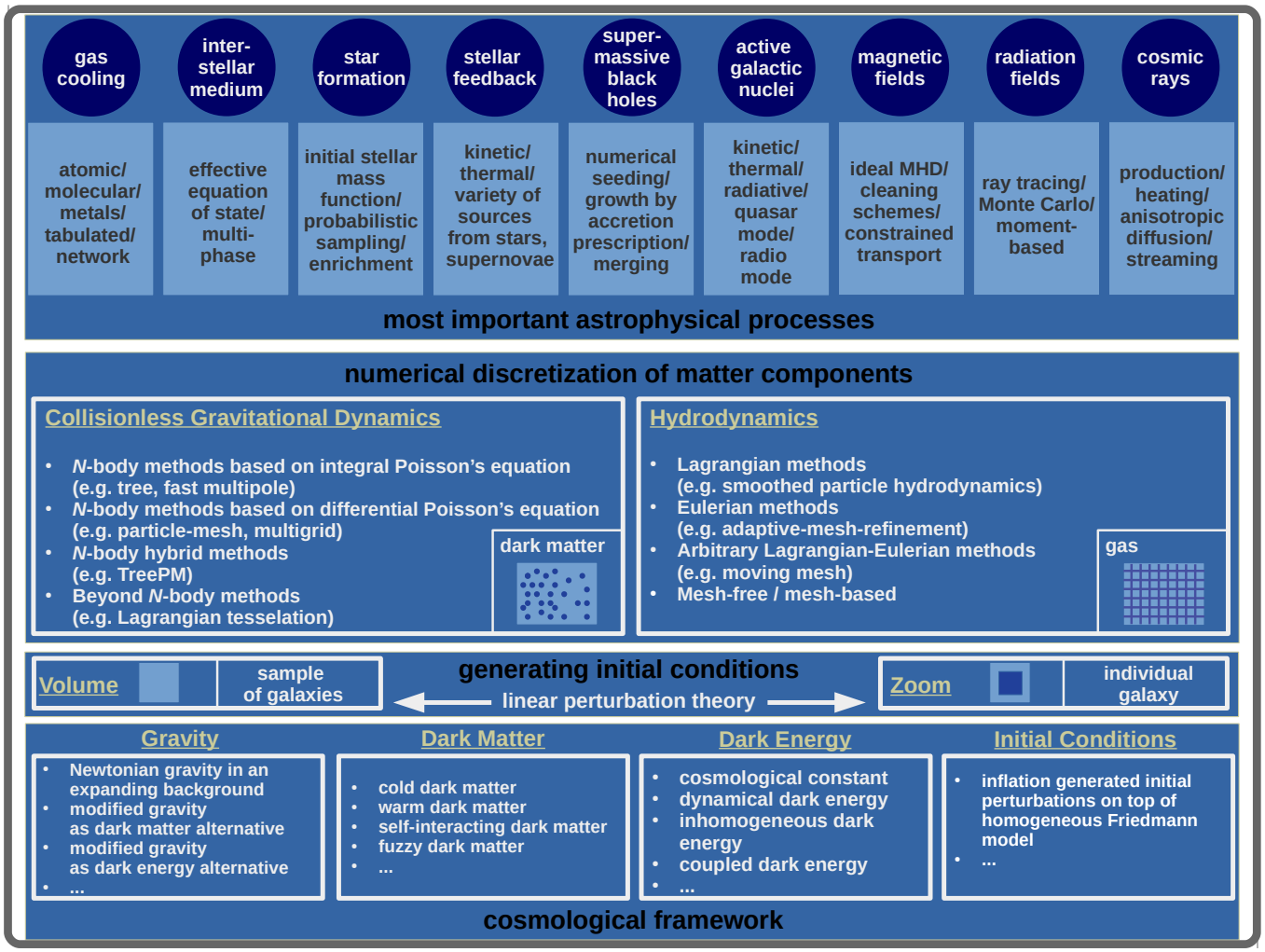


Figure 1. Overview of the key ingredients of cosmological simulations (adapted from Vogelsberger et al. (2019)). Simulations are performed within a given cosmological framework and start from specific initial conditions. This framework includes physical models for gravity, dark matter, dark energy, and baryonic processes such as star formation, stellar feedback, and AGN feedback.

Redshift (z)	α_s	β_s	α	β	Scatter (dex)
6.0	0.091	-9.139	1.091	1.771	0.182
5.0	0.082	-9.257	1.082	1.563	0.218
4.0	0.079	-9.422	1.079	1.373	0.251
3.0	0.071	-9.612	1.071	1.093	0.278
2.0	0.088	-10.065	1.088	0.812	0.313
1.0	-0.004	-9.654	0.996	0.304	0.344
0.5	-0.095	-9.082	0.905	-0.307	0.385
0.0	-0.136	-8.945	0.864	-0.303	0.431

Table 1. Best-fit parameters of the star-forming main sequence (SFMS) shown in 2 in the IllustrisTNG50 simulation. The relation is defined as $\log_{10}\mathrm{sSFR} = \alpha_s(z)\,\log_{10}(M_\star) + \beta_s(z)$, where $\alpha_s = [\alpha(z) - 1]$ is the slope and $\beta_s(z) = \beta - 10\alpha$ is the normalization at $10^{10}\mathrm{M}_\odot$. Scatter is the root-mean-square (RMS) dispersion of the residuals in log–log space, measured in dex. Fits include only main-sequence galaxies ($10^7 < M_\star < 10^{10}$) and exclude quenched systems.

4 RESULTS & ANALYSIS

4.1 The Star-forming Main Sequence Across Cosmic Time

We begin by characterizing the evolution of the star-forming main sequence (SFMS) from $z = 6$ to $z = 0$ using the IllustrisTNG50 simulation. Figure 2 shows the specific star formation rate (sSFR≡SFR/$M_\star$) as a function of stellar mass for galaxies with $M_\star \geq 10^7\,\mathrm{M}_\odot$ at eight redshift snapshots. Each panel displays individual galaxies as black points, with both axes plotted on logarithmic scales. The red line in each panel represents the best-fit linear relation in log-log space, obtained by fitting only galaxies in the mass range $10^7 \leq M_\star/\mathrm{M}_\odot \leq 10^{10}$ with sSFR $> 10^{-12}\,\mathrm{yr}^{-1}$, thereby excluding low-mass systems, massive quenched galaxies, and extreme outliers.

The SFMS exhibits clear redshift evolution. At $z = 6$, the relation is characterized by $\log_{10}(\mathrm{sSFR}) \approx -8.5$ for a typical main-sequence galaxy, with a relatively tight scatter of $\sigma = 0.18$ dex (Table 1). Nearly all galaxies at this epoch lie near or above the fitted relation, indicating vigorous, widespread star formation. The slope of the sSFR-$M_\star$ relation at $z = 6$ is $\alpha \approx 1.09$, meaning that sSFR increases slightly with stellar mass, or equivalently, that more massive galaxies form stars somewhat more efficiently relative to their existing stellar mass.

As cosmic time progresses, the normalization of the SFMS systematically declines. By $z = 2$, typical sSFR values have dropped to approximately $10^{-9.5}\,\mathrm{yr}^{-1}$, and by $z = 0$ they decline by nearly two orders of magnitude to $10^{-10.5}\,\mathrm{yr}^{-1}$. This evolution could be due the cosmic decline in cold gas availability as the universe ages and

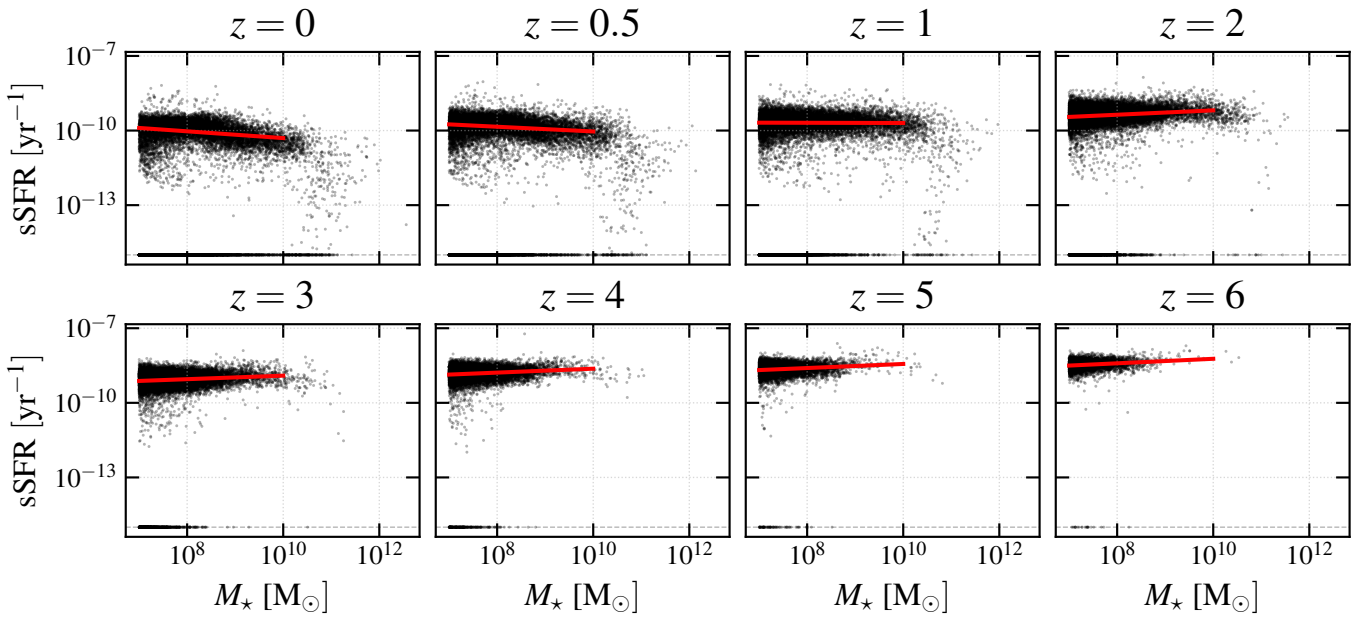


Figure 2. Specific star-formation rate (sSFR ≡ SFR/$M_\star$) versus stellar mass ($M_\star$) for galaxies in the IllustrisTNG50 simulation at redshifts z = 0–6. Each panel shows individual galaxies with $M_\star \geq 10^7$ M$_\odot$, and both axes are plotted on logarithmic scales. The red line in each subplot represents the best-fit relation to the star-forming main sequence obtained from a linear regression in log space. The fit is performed only on galaxies with $10^7 < M_\star/\mathrm{M}_\odot < 10^{10}$ and with sSFR $> 10^{-12}$ yr^{-1}, thereby excluding low-mass systems, massive quenched galaxies, and sSFR outliers. The resulting best-fit parameters (slope, normalization, and scatter) for each redshift are provided in Table 1. The decreasing normalization of the sSFR–$M_\star$ relation toward low redshift reflects the decline of cosmic star-formation activity over time.

the increasing efficiency of feedback processes that heat and expel gas. The slope also evolves, approaching $\alpha \approx 1$ at $z \approx 1$ (Table 1), which corresponds to $\alpha_s \approx 0$ in the sSFR formulation. At this epoch, sSFR becomes nearly independent of stellar mass, and the $\log_{10}(\mathrm{sSFR}) - \log_{10}(\mathrm{M}_\star)$ relation appears nearly horizontal, consistent with observational findings (Rodighiero et al. 2011). At later times, the slope decreases further, reaching $\alpha \approx 0.86$ at $z = 0$, indicating that lower-mass galaxies maintain higher sSFR compared to their more massive counterparts.

The scatter around the SFMS increases steadily from $\alpha = 0.18$ dex at $z = 6$ to $\alpha = 0.43$ dex at $z = 0$. This broadening reflects the growing diversity of galaxy evolutionary states. We can identify three distinct populations in Figure 2:

Main-sequence galaxies constitute the dense horizontal ridge visible in each panel. These systems represent the majority of the star-forming population at all epochs. Galaxies on the main sequence exhibit gradually lower specific star formation rates across cosmic time.

Starburst galaxies lie above the main sequence, typically with sSFR values one order of magnitude or more above the best-fit line (sSFR $\geq 10^{-9}$ yr^{-1} at low redshift). These systems are more prominent at $z \leq 3$ and represent galaxies undergoing enhanced star formation, likely triggered by gas-rich mergers, interactions, or other processes that funnel large amounts of gas into the galaxy center. Starbursts are inherently short-lived because they rapidly consume their gas reservoirs (the gas depletion time, $\tau_{\mathrm{dep}} \approx M_{\mathrm{gas}}/\mathrm{SFR}$; Cimatti et al. (2020)).

Quiescent galaxies appear as a growing population below the main sequence at sSFR $\lesssim 10^{-11}$ yr^{-1}. This population is nearly absent at $z \gtrsim 4$ but becomes increasingly prominent at $z \lesssim 1$. These are galaxies that have been quenched, meaning that their star formation has been suppressed or halted entirely, either through gas exhaustion or feedback processes.

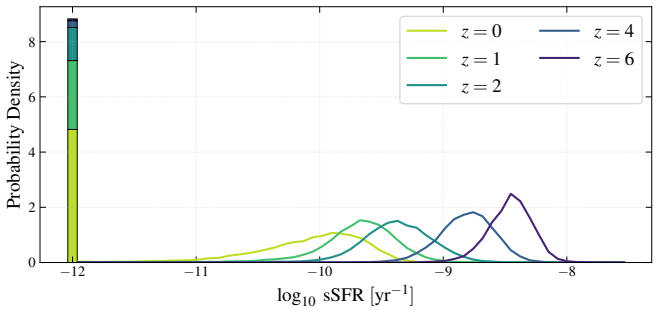


Figure 3. Probability density functions of $\log_{10}$(sSFR) for galaxies with $M_\star \geq 10^7$ M$_\odot$ in IllustrisTNG50 at z = 0, 1, 2, 4, and 6. PDFs are normalized so that the integral over all sSFR (including the stacked bar) equals unity at each redshift. The stacked bar at $\log_{10}(\mathrm{sSFR}) = -15$ represents galaxies with sSFR $\leq 10^{-15}$ yr^{-1} (effectively zero star formation). The fraction of quiescent galaxies grows from ~0% at $z = 6$ to ~50% at $z = 0$, while the peak of the star-forming distribution shifts from $\log_{10}(\mathrm{sSFR}) \approx -8.5$ to ≈ -10.5.

The best-fit parameters for the SFMS at each redshift are provided in Table 1. Our simulated SFMS shows good agreement with the observational synthesis of Speagle et al. (2014), particularly in the systematic decline of normalization and the transition to $\alpha \approx 1$ at $z \sim 1 - 2$.

4.2 The Distribution of the Specific Star Formation Rates

To better understand the evolution of the star-forming and quiescent populations, we examine the probability distribution functions (PDFs) of sSFR. Figures 3 and 4 present complementary views of these distributions across cosmic time. Figure 3 shows the complete sSFR distribution, including galaxies with effectively zero star for-

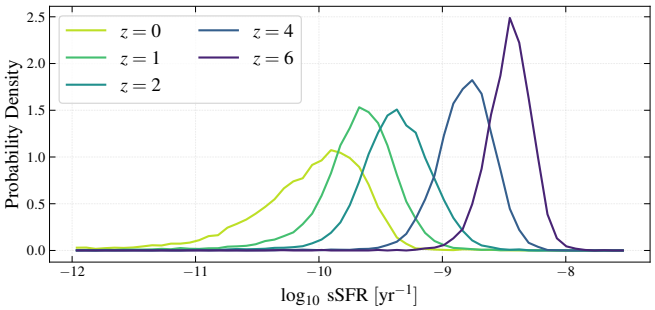


Figure 4. Probability density functions of $\log_{10}$(sSFR) for star-forming galaxies (sSFR $> 10^{-15}$ yr^{-1}) with $M_\star \geq 10^7$ M$_\odot$ in IllustrisTNG50. PDFs are normalized to integrate to unity over the displayed range at each redshift. The distributions shift systematically toward lower sSFR from $z = 6$ to $z = 0$, and the scatter increases, reflecting both the cosmic decline in star formation efficiency and the growing diversity of star formation histories.

mation. Galaxies that have very low sSFR (sSFR $\leq 10^{-15}$ yr^{-1}) are binned together and represented as a stacked bar at $\log_{10}$(sSFR) $= -15$. The height of each colored segment in this bar quantifies the fraction of galaxies at that redshift that are quenched. At $z = 6$, this fraction is negligible, and nearly every galaxy with $M_\star \geq 10^7$ M$_\odot$ is actively forming stars. By $z = 0$, approximately 50% of galaxies in our sample have ceased star formation, reflecting the dramatic buildup of the quiescent population.

The continuous curves next to the bar show the distribution of actively star-forming galaxies. At $z = 6$, the peak lies at $\log_{10}$(sSFR) ≈ -8.5, and shifts steadily leftward (toward lower sSFR) as the universe evolves, reaching $\log_{10}$(sSFR) ≈ -10.5 by $z = 0$. The shape of the distribution also changes: at high redshift, it is sharply peaked, indicating that most galaxies form stars at similar specific rates. By low redshift, the distribution has broadened and developed a pronounced tail extending toward very low sSFR.

Figure 4 presents the same data but excludes galaxies with sSFR $\leq 10^{-15}$ yr^{-1}, allowing us to focus exclusively on the star-forming population. The width of each curve represents the intrinsic scatter in the SFMS. At $z = 6$, the distribution is narrow, consistent with the small scatter ($\alpha = 0.18$ dex) found in Figure 2. As redshift decreases, the distributions broaden, and by $z = 0$ the PDF is substantially wider, with a clear tail toward lower sSFR. This broadening indicates that the star-forming population itself becomes more diverse over time, with some galaxies maintaining high sSFR while others experience declining star formation efficiency.

4.3 The Coevolving Relationship between Black Hole Mass and Stellar Mass

We now turn to the relationship between black hole mass and stellar mass, examining how this scaling relation evolves across cosmic time. Figure 5 displays $M_{\rm BH}$ versus $M_\star$ for galaxies in IllustrisTNG50 at the same eight redshift snapshots used in Figure 2. Black points show individual galaxies with $M_\star \geq 10^7$ M$_\odot$, and the orange line represents the best-fit relation obtained by fitting galaxies with $M_{\rm BH} \geq 10^6$ M$_\odot$ in log-log space. At all redshifts, the simulated galaxies exhibit a tight correlation between $M_{\rm BH}$ and $M_\star$. This correlation tightens at lower redshift as black holes grow through accretion and mergers. At $z = 6$, black hole masses cluster strongly near 10^6 M$_\odot$, which is the seed mass used in the IllustrisTNG seeding prescription, with only a small population of more massive black holes that have experienced early growth. The slope of the $M_{\rm BH} - M_\star$ relation at this epoch is shallow ($p = 0.26$, Table 2), and the normalization is low: at $M_\star = 10^{10}$ M$_\odot$, the typical black hole mass is $\log_{10}(M_{\rm BH}/{\rm M}_\odot)6.6$.

Redshift (z)	p	q	Scatter (dex)
6.0	0.260	6.561	0.106
5.0	0.332	6.682	0.131
4.0	0.453	6.877	0.179
3.0	0.599	7.115	0.203
2.0	0.727	7.331	0.206
1.0	0.794	7.475	0.204
0.5	0.799	7.533	0.200
0.0	0.787	7.569	0.202

Table 2. Best-fit parameters of the black hole-stellar mass relation (Eq. 1) in the IllustrisTNG50 simulation. p is the slope, q is the normalization (the expected $\log M_{\rm BH}$ at $M_\star = 10^{10}$ M$_\odot$), and 'Scatter' denotes the rms dispersion (in dex) about the best-fit line.

As cosmic time progresses, the black hole population evolves dramatically. The distribution broadens, developing a pronounced high-mass tail as an increasing fraction of black holes grow well beyond their initial seed mass. By $z = 0$, the slope has steepened to $p = 0.79$, approaching the nearly linear relationship ($p \approx 1$) observed in some local samples. The normalization has also increased substantially: by $z = 0$, a galaxy with $M_\star = 10^{10}$ M$_\odot$ typically hosts a black hole with $\log_{10}(M_{\rm BH}/{\rm M}_\odot) \approx 7.6$, which is an increase of one order of magnitude relative to $z = 6$. The scatter around the relation remains relatively constant at $\sigma \approx 0.2$ dex across all redshifts (Table 2), indicating that the $M_{\rm BH} - M_\star$ correlation is maintained as both populations grow.

We compare our simulated relation to two observational benchmarks. The blue dashed line in Figure 5 shows the classical $M_{\rm BH} - M_{\rm bulge}$ relation from Kormendy & Ho (2013), derived from local galaxies ($z \approx 0$). At low redshift ($z = 0$ and $z = 0.5$), our simulated relation shows good agreement with the Kormendy & Ho (2013) fit. This is expected because the TNG AGN feedback model was calibrated in part to reproduce local black hole-galaxy scaling relations. The agreement suggests that, for the massive galaxies dominating the high-mass end of our sample, total stellar mass and bulge mass are approximately equivalent, as most of the stellar mass resides in the central bulge component. At high redshift ($z \gtrsim 2$), however, our simulated relation deviates systematically from the (Kormendy & Ho 2013) fit. Black hole masses cluster near the seed mass of 10^6 M$_\odot$, producing a concentrated distribution at the lower left of each high-redshift panel. This clustering is because the IllustrisTNG model does not simulate the physical formation process of black holes (e.g., from Population III stellar remnants or direct collapse), but instead seeds black holes with a fixed mass when dark matter halos exceed a threshold mass. At high redshift, most galaxies are young, and their black holes have not had sufficient time to grow through accretion or mergers. As a result, the slope of the $M_{\rm BH} - M_\star$ relation is shallow, and most systems remain close to the seed mass.

Red points in the $z = 0$ and $z = 0.5$ panels show observational measurements from Reines & Volonteri (2015). Their sample comprises 262 broad-line AGN at $z < 0.055$ and 79 galaxies with dynamical black hole mass measurements from the Sloan Digital Sky Survey (SDSS). Black hole masses in their sample are estimated using virial techniques based on broad emission line widths and AGN luminosities. The observed galaxies exhibit a correlation between $M_{\rm BH}$ and $M_\star$, confirming the fundamental connection between black holes and their hosts. However, the Reines & Volonteri (2015) data show substantial scatter and do not fall precisely on the same sequence as our best-fit relation. There may be several reasons for this offset that we discuss in Section 5.

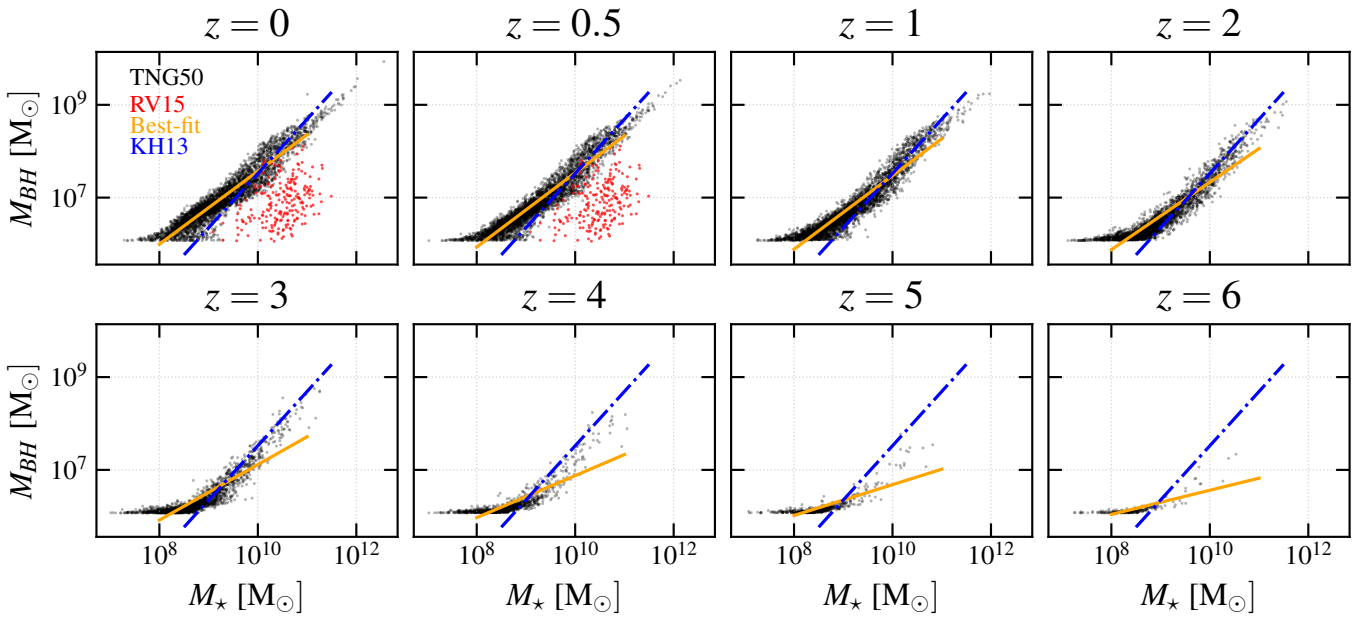


Figure 5. Black hole mass ($M_{\rm BH}$) versus stellar mass ($M_\star$) for galaxies in IllustrisTNG50 at $z = 0$–6. Black points show individual galaxies with $M_\star \geq 10^7$ M$_\odot$; orange lines show best-fit relations (see Table 2); The blue dashed line shows the Kormendy & Ho (2013) $M_{\rm BH}$–$M_{\rm bulge}$ relation for local galaxies. Red points ($z = 0, 0.5$ panels only) show observations from Reines & Volonteri (2015): 262 broad-line AGN at $z < 0.055$ with virial black hole mass estimates from SDSS. The simulation relation agrees well with Kormendy & Ho (2013) at $z \lesssim 1$ but deviates at $z \gtrsim 2$, where black holes cluster near the 10^6 M$_\odot$ seed mass.

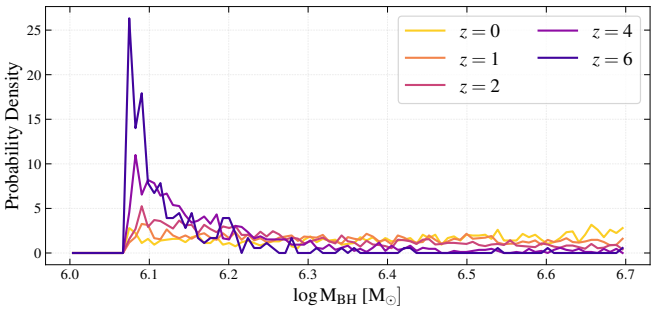


Figure 6. Probability density functions of $\log_{10}(M_{\rm BH})$ for galaxies with $M_\star \geq 10^7$ M$_\odot$ in IllustrisTNG50. PDFs are normalized to integrate to unity at each redshift. The sharp peak at $\log_{10}(M_{\rm BH}/{\rm M}_\odot) \approx 6.1$ corresponds to the simulation seed mass (8×10^5 M$_\odot/h \approx 1.2 \times 10^6$ M$_\odot$). The peak shifts slightly to higher masses and a prominent high-mass tail develops by $z = 0$, reflecting cumulative black hole growth through accretion and mergers.

Figure 6 shows the probability distribution function (PDF) of black hole mass at five representative redshifts. This figure complements Figure 5 by illustrating how the entire black hole population builds up over cosmic time, rather than just the instantaneous $M_{\rm BH} - M_\star$ correlation. At $z = 6$, the black hole mass distribution sharply peaks at $\log_{10}(M_{\rm BH}/{\rm M}_\odot) \approx 6.1$, corresponding to the simulation seed mass of $\sim 1.3 \times 10^6$ M$_\odot$ or 8×10^5 M$_\odot$/h (Nelson et al. 2021). As redshift decreases, the peak shifts gradually toward higher masses as black holes accrete gas and merge. By $z = 0$, the peak has moved to $\log_{10}(M_{\rm BH}/{\rm M}_\odot) \approx 6.3 - 6.5$. The evolution of black hole mass directly reflects the evolution of the $M_{\rm BH} - M_\star$ scaling relation documented in Figure 5. The sharp seed-mass peak at high redshift corresponds to the lower envelope of the $M_{\rm BH} - M_\star$ relation. The growing high-mass tail corresponds to the upward shift in normalization, and as galaxies grow in stellar mass and their black holes accrete efficiently, the population moves to higher $M_{\rm BH}$ at fixed $M_\star$.

4.4 Black Hole Accretion Rates and Star Formation Activity

After characterizing the black hole population and its mass distribution, we now examine black hole accretion activity and its connection to star formation. Figure 7 shows the black hole accretion rate ($\dot{M}$) as a function of stellar mass for galaxies in IllustrisTNG50, with points color-coded by specific star formation rate (sSFR). A floor value of $\dot{M} = 10^{-13}$ M$_\odot$ yr^{-1} is applied, so black holes with lower accretion rates are not displayed.

At all redshifts, a well-defined sequence of actively accreting black holes is visible, with $\dot{M}$ increasing with stellar mass. At $z = 6$, the accretion rate sequence is tight and nearly monotonic, and the trend is that more massive galaxies host black holes with higher accretion rates. Typical accretion rates at this epoch are $\dot{M} \sim 10^{-3}$ to 10^{-2} M$_\odot$ yr^{-1} for galaxies with $M_\star \sim 10^{10}$ M$_\odot$. The color-coding shows that these actively accreting black holes reside almost exclusively in star-forming galaxies (orange to red colors, indicating $\log_{10}({\rm sSFR}) \gtrsim -9$). As redshift decreases, the accretion rate distribution undergoes significant changes. By $z \approx 2 - 3$, the sequence begins to broaden, with increasing scatter in $\dot{M}$ at fixed $M_\star$. By $z \lesssim 1$, we see two populations emerging. The actively star-forming galaxies indicated by blue dots continue to accrete onto their black holes at moderate to high levels ($10^{-5} \leq \log_{10}(\dot{M}) \leq 10^{-2}$), while the quenched systems with low sSFR, indicated by orange and red dots, form a sequence of diverse stellar masses and accretion rates.

Figure 8 quantifies this evolution by showing the probability distribution function of black hole accretion rates at five representative redshifts. At $z = 6 - 4$, the PDF peaks at $\log_{10}(\dot{M}) \approx -2.5$ to -3, corresponding to $\dot{M} \sim 10^{-3}$ to $10^{-2.5}$ M$_\odot$ yr^{-1}. As cosmic time progresses, the peak shifts systematically toward lower accretion rates. By $z = 0$, the peak has moved to $\log_{10}(\dot{M}) \approx -4$ to -4.5 M$_\odot$ yr^{-1}), representing a decline of approximately 1.5 orders of magnitude.

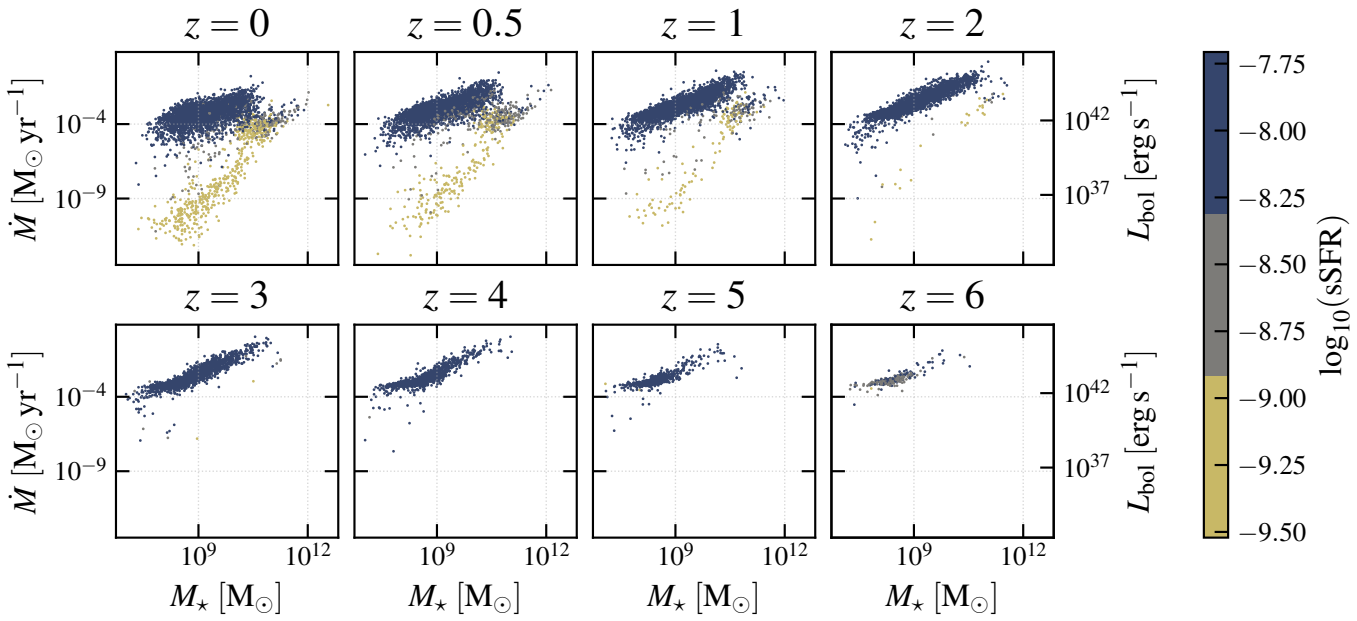


Figure 7. Black hole accretion rate ($\dot{M}$) versus stellar mass ($M_\star$) for galaxies in IllustrisTNG50 at z = 0–6. Points are color-coded by $\log_{10}$(sSFR). Only black holes with $\dot{M} > 10^{-13}$ M$_\odot$ yr^{-1} are shown. At high redshift, actively accreting black holes form a tight sequence correlated with stellar mass and reside in star-forming galaxies (blue colors indicate high sSFR; orange/red indicate low sSFR). At $z \lesssim 1$, the distribution broadens significantly, with most massive galaxies hosting low-accretion black holes in quiescent systems.

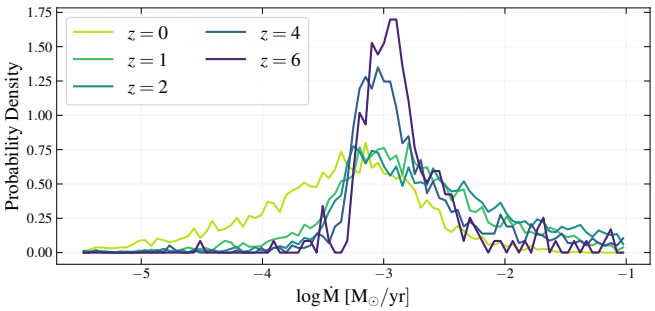


Figure 8. Probability density functions of $\log_{10}(\dot{M})$ for black holes in galaxies with $M_\star \geq 10^7$ M$_\odot$ in IllustrisTNG50. PDFs are normalized to integrate to unity at each redshift. The peak shifts from $\log_{10}(\dot{M}/\mathrm{M}_\odot\ \mathrm{yr}^{-1}) \approx -2.5$ at $z = 6$ to ≈ -4.5 at $z = 0$ (∼1.5 dex decline), and the distribution broadens, reflecting both the cosmic decline in gas supply and the growing population of quiescent systems with minimal black hole activity.

4.5 Black Hole Growth Efficiency and Mass Ratios

Figure 9 explores the relationship between black hole growth efficiency and the galaxy-to-black-hole mass ratio by plotting the specific accretion rate ($\dot{M}/M_{\rm BH}$) as a function of $M_\star/M_{\rm BH}$. Points are color-coded by star formation rate (SFR, not sSFR) to indicate the galaxy's star-forming state. At $z = 6-4$, galaxies cluster in a relatively narrow region with $M_\star/M_{\rm BH} \sim 10^2 to 10^3$, indicating that black holes and their host galaxies maintain an approximately constant mass ratio during early cosmic times. As redshift decreases, the distribution broadens significantly. By $z \lesssim 1$, galaxies span a much wider range of $M_\star/M_{\rm BH}$ ratios, from $\sim 10^1$ to beyond 10^3, reflecting increasing diversity in the evolutionary states of the black hole and galaxy populations. The scatter in specific accretion rates ($\dot{M}/M_{\rm BH}$) at fixed $M_\star/M_{\rm BH}$ also increases toward lower redshift. At $z = 0$, galaxies with similar stellar-to-black-hole mass ratios can have specific accretion rates differing by several orders of magnitude, suggesting that $M_\star/M_{\rm BH}$ alone does not uniquely determine the black hole growth rate. The color-coding reveals some correlation between star formation activity and black hole accretion: galaxies with higher SFR (blue colors) tend to cluster at the high-end tail of the specific accretion rates, while those with lower SFR (orange to red colors) are distributed across the parameter space.

4.6 Comparing IllustrisTNG100 and Illustris

We now look at individual galaxy histories to explore how gas depletion, star formation, and black hole accretion are coupled on galaxy-by-galaxy timescales. We begin by comparing gas mass evolution in IllustrisTNG100 with the older Illustris simulation. Figure 10 shows the gas mass within the stellar half-mass radius (the radius enclosing 50% of the galaxy's stellar mass) as a function of cosmic time for randomly selected galaxies in three stellar mass bins: $M_\star \sim 10^9$ M$_\odot$ (left), $M_\star \sim 10^{10}$ M$_\odot$ (middle), and $M_\star \sim 10^{11}$ M$_\odot$ (right). These masses refer to the final stellar mass at $z = 0$. Blue solid lines represent galaxies from IllustrisTNG100, while black dashed lines show their "sibling" galaxies in Illustris. The two simulations use the same initial conditions (identical phases of density fluctuations, adjusted for the updated cosmology), meaning that galaxies form in the same locations and experience similar large-scale environments.

For low-mass galaxies ($M_\star \sim 10^9$ M$_\odot$, left panel), both simulations show relatively smooth gas mass evolution over cosmic time. Gas masses remain in the range $\log_{10}(M_{\rm gas}/\mathrm{M}_\odot) \sim 7.5-10$ throughout most of cosmic history. However, the TNG100 galaxy maintains gas masses approximately one order of magnitude lower than its Illustris sibling for much of its history. Nevertheless, both galaxies retain substantial gas reservoirs and continue forming stars throughout cosmic time. In the intermediate-mass bin ($M_\star \sim 10^{10}$ M$_\odot$, middle panel), the gas mass histories become more variable, with episodic fluctuations. The TNG100 galaxy exhibits somewhat lower

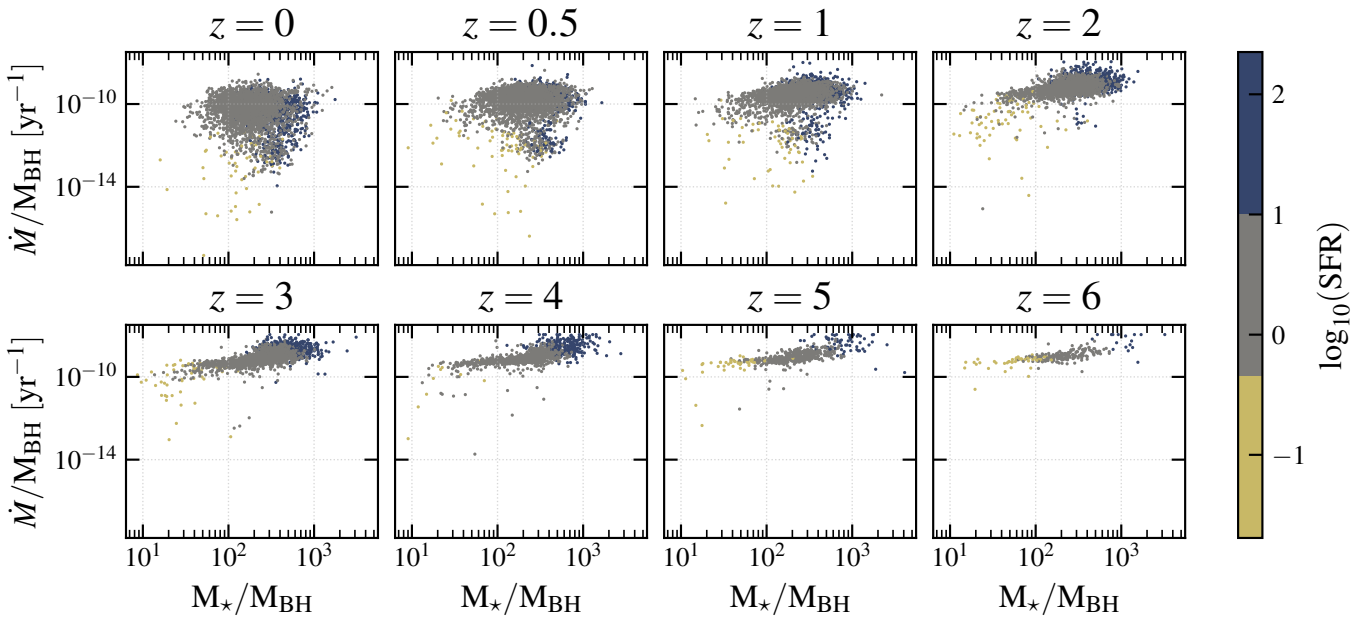


Figure 9. Specific black hole accretion rate ($\dot{M}/M_{\rm BH}$) versus stellar-to-black-hole mass ratio ($M_\star/M_{\rm BH}$) for galaxies with $M_\star \geq 10^7$ M$_\odot$ in IllustrisTNG50. Points are color-coded by $\log_{10}$(SFR), where blue indicates high SFR ($\gtrsim 10$ M$_\odot$ yr^{-1}) and orange/red indicates low SFR ($\lesssim 1$ M$_\odot$ yr^{-1}). At $z \gtrsim 4$, galaxies cluster narrowly at $M_\star/M_{\rm BH} \sim 10^2$–$10^3$. By $z \lesssim 1$, the distribution broadens significantly, spanning $M_\star/M_{\rm BH} \sim 10^1$–$10^3$ with diverse specific accretion rates.

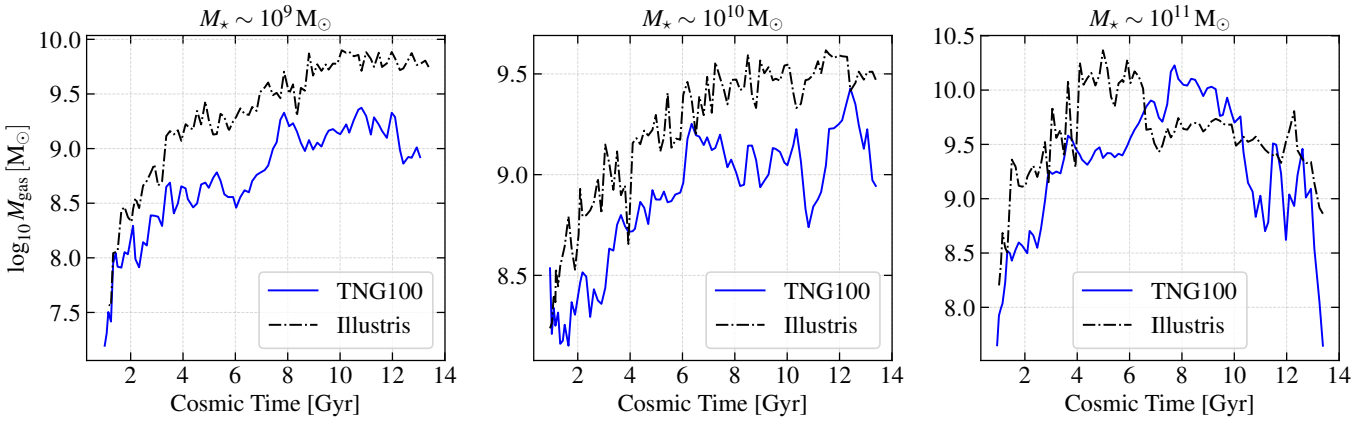


Figure 10. Gas mass as a function of cosmic time for three different mass bins for galaxies in the IllustrisTNG100 simulation and their sibling galaxies in the older Illustris simulation. The blue solid line are galaxies from the IllustrisTNG100 simulation, chosen at random, while the black dashed line is the sibling galaxies in the Illustris simulation. The left subplot shows two galaxies in the $\sim 10^9$ M$_\odot$ mass range, middle subplot shows two galaxies in the $\sim 10^{10}$ M$_\odot$ mass range, while the rightmost subplot shows massive galaxies in the $\sim 10^{11}$ M$_\odot$ mass range. The y-axis represents the log scale gas mass in the stellar half-mass radius of the galaxy, measured in solar masses. The x-axis represents cosmic time in gigayears, converted from redshift using the `astropy.cosmology` Python package

gas masses at late times ($t \gtrsim 9$ Gyr, $z \lesssim 0.5$) compared to its Illustris counterpart. The most dramatic differences appear in the massive galaxy bin ($M_\star \sim 10^{11}$ M$_\odot$, right panel). Here, both simulations exhibit sharp drops in gas mass at intermediate cosmic times ($t \sim 6 - 10$ Gyr, corresponding to $z \sim 1 - 0.5$), but the TNG100 galaxy experiences a more abrupt and complete gas depletion event.

4.7 Tracing Galaxy Evolution

Figure 11 extends the analysis of Figure 10 by tracking three quantities simultaneously for randomly selected galaxies in each mass bin: gas mass within the stellar half-mass radius (top row), specific star formation rate (middle row), and black hole accretion rate (bottom row). Three galaxies are shown per mass bin (red, blue, and black curves), chosen randomly from systems with final stellar mass $M_\star \sim 10^9$, 10^10, and 10^{11} M$_\odot$ at $z = 0$. All quantities are from IllustrisTNG100 and are plotted as functions of cosmic time measured in gigayears.

Low-mass galaxies ($M_\star \sim 10^9$ M$_\odot$, left column): The three quantities evolve relatively smoothly throughout cosmic history. Gas mass evolves in the range $\log_{10}(M_{\rm gas}/{\rm M}_\odot) \sim 7.5 - 8.5$ with no dramatic depletion events. The specific star formation rate remains high, $\log_{10}({\rm sSFR}) \gtrsim -10\,{\rm yr}^{-1}$, throughout most of cosmic time, though the three galaxies exhibit gradually lower sSFR by $t = 14$ Gyr. Black hole accretion rates show occasional brief dips to lower values. One galaxy (red curve) experiences a dramatic drop in black hole accre-

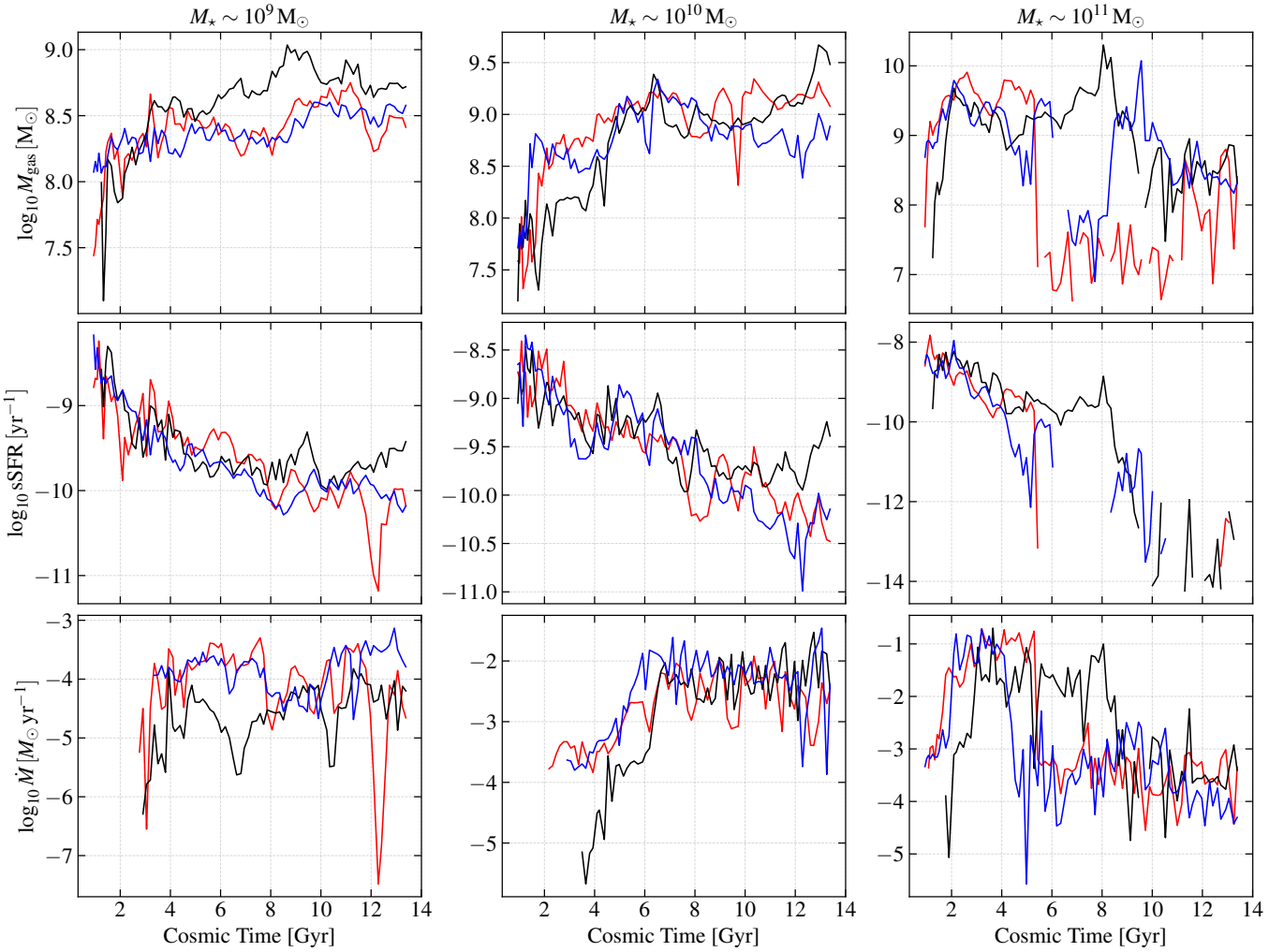


Figure 11. Cosmic evolution of gas mass, star formation activity, and black hole accretion for representative galaxies across three mass bins from the IllustrisTNG100 simulation. Each column corresponds to a different stellar mass bin: $\sim 10^9\,\mathrm{M}_\odot$ (left),$\sim 10^{10}\,\mathrm{M}_\odot$ (middle), and $\sim 10^{11}\,\mathrm{M}_\odot$ (right). For each mass bin, three randomly selected galaxies are shown (red, blue, and black lines). The top row displays the log-scale gas mass within the stellar half-mass radius ($M_{\rm gas}$) measured in solar masses, the middle row shows the log-scale specific star formation rate in the galaxy half-mass radius (sSFR) in yr^{-1} units, and the bottom row presents the log-scale black hole accretion rate ($\dot{M}$) measured in solar masses per year. All quantities are plotted as a function of cosmic time measured in gigayears.

tion rate at $t \sim 12$ Gyr, with $\dot{M}$ declining by more than two orders of magnitude. Interestingly, this event is not accompanied by a corresponding drop in gas mass or sSFR.

Intermediate-mass galaxies ($M_\star \sim 10^{10}\,\mathrm{M}_\odot$, middle column): The histories are more dynamic. Gas masses show episodic variability, with several galaxies experiencing drops of ~ 0.5 dex, but overall it follows a similar growth as the low mass bin to the left. We see a gradual decrease in specific star formation ($\log_{10}(\mathrm{sSFR}) \approx -8.5$ at $t \sim 2$ Gyr to ≈ -10.5 by $t \sim 14$ Gyr) often overlapping with an increase in black hole accretion, and both are preceded or accompanied by increases in gas mass.

Massive galaxies ($M_\star \sim 10^{11}\,\mathrm{M}_\odot$, right column): The behavior changes qualitatively. All three galaxies exhibit sharp, nearly simultaneous drops in all three quantities at cosmic times $t \gtrsim 5$ Gyr (corresponding to $z \lesssim 1$). Prior to these transitions, the galaxies maintain high gas masses ($\log_{10}(M_{\rm gas}/\mathrm{M}_\odot) \gtrsim 9$), elevated sSFR ($\log_{10}(\mathrm{sSFR}) \sim -9.5$ to $-10\,\mathrm{yr}^{-1}$, and vigorous black hole accretion ($\log_{10}(\dot{M}) \sim -2$ to -3. Following the transition, conditions change dramatically. Gas masses plummet by more than one order of magnitude to ($\log_{10}(M_{\rm gas}/\mathrm{M}_\odot) \sim 8.5 - 9$). Specific star formation rates drop to $\log_{10}(\mathrm{sSFR}) \lesssim -12\,\mathrm{yr}^{-1}$. Black hole accretion rates decline by 2-3 orders of magnitude to $\dot{M} \lesssim 10^{-5}$ to $10^{-4}\,\mathrm{M}_\odot\,\mathrm{yr}^{-1}$.

4.8 Quantifying Temporal Correlations

Figure 12 quantifies the temporal relationship between gas depletion events and the responses in star formation rate and black hole accretion rate. This lag analysis addresses a key question raised by Figure 11: when gas is suddenly removed from a galaxy, do SFR and $\dot{M}$ respond instantaneously, or is there a measurable delay?

We focus on massive galaxies ($M_\star \sim 10^{11}\,\mathrm{M}_\odot$ at $z = 0$) in IllustrisTNG100 and identify all intervals where the gas mass within the stellar half-mass radius drops by at least a factor of 10 (1 dex) between consecutive snapshots. For each such "gas depletion event," we compute the median change in black hole accretion rate ($\Delta \log_{10} \dot{M}$, top panel) and star formation rate ($\Delta \log_{10}$ SFR, bottom panel) as a function of time lag. The time lag is measured relative to the gas depletion event, with negative lags indicating times before the event and pos-

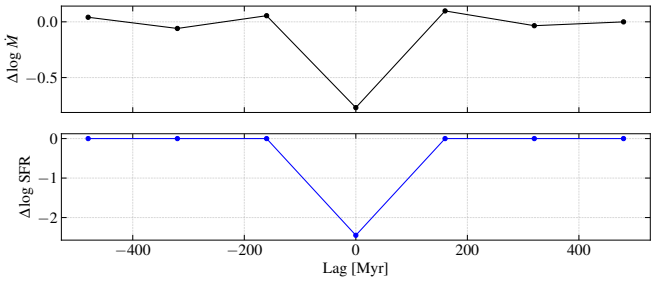


Figure 12. Temporal lag analysis for massive galaxies ($M_\star \sim 10^{11}$ M$_\odot$ at $z = 0$) in IllustrisTNG100. Top: median change in black hole accretion rate ($\Delta \log_{10} \dot{M}$). Bottom: median change in star formation rate ($\Delta \log_{10}$SFR). Both are measured relative to gas mass depletion events ($\geq$1 dex drop in $M_{\rm gas}$ within $r_{1/2}$ between consecutive snapshots). Time lag spans ±3 snapshots (∼±480 Myr; snapshot spacing ∼160 Myr). Both quantities reach minimum values (largest drops) at zero lag with symmetric responses, indicating that SFR and $\dot{M}$ respond to gas removal on timescales shorter than the simulation's temporal resolution, consistent with rapid AGN-driven feedback.

itive lags indicating times after. We span ±3 snapshots around each event, corresponding to approximately ±480 Myr given the ∼ 160 Myr spacing between snapshots.

Both panels exhibit their minimum values corresponding to the largest drops in $\dot{M}$ and SFR at zero lag. The top panel shows that black hole accretion rates decline by $\Delta \log_{10} \dot{M} \simeq -0.5$ dex coincident with the gas depletion event. The bottom panel indicates that star formation rates drop by $\Delta \log_{10}$ SFR $\simeq -2$ dex at the same time. The responses are symmetric around zero lag, with no systematic offset toward positive or negative lags. The curves rise toward zero at both positive and negative lags, indicating that the changes in $\dot{M}$ and SFR are specifically associated with the gas depletion event. This result indicates that, within the ∼ 160 Myr temporal resolution of the simulation, star formation and black hole accretion respond essentially instantaneously to the removal of cold gas. Any intrinsic lag between gas removal and the suppression of SFR or $\dot{M}$ is shorter than the time between consecutive snapshots.

5 DISCUSSIONS AND CONCLUSIONS

5.1 Summary of Results

This study presents a view of the joint evolution of galaxies and their central supermassive black holes accross cosmic time using the IllustrisTNG50 and IllustrisTNG100 simulations. Our analysis reveals the interconnected evolutionary processes that shape both the star-forming main sequence and the black hole population over cosmic time. Our main findings are:

Star-forming main sequence evolution: The specific star formation rate of galaxies in IllustrisTNG50 declines by nearly two orders of magnitude from $z = 6$ to $z = 0$, reflecting the cosmic decline in gas availability and the increasing importance of feedback processes. The normalization of the sSFR − M$_\star$ relation evolves from $\log_{10}$(sSFR) ≈ -8.5 at $z = 6$ to ≈ -10.5 at $z = 0$ for typical main-sequence galaxies. The scatter increases from $\sigma \approx 0.18$ dex at high redshift to $\sigma \approx 0.43$ dex at $z = 0$, driven by the emergence of a quiescent population.

Black hole mass scaling: Simulated galaxies in IllustrisTNG50 exhibit at tight $M_{BH} - M_\star$ correlation at all redshifts, with both slope and normalization evolving systematically toward higher black hole masses at fixed stellar mass. The slope increases from $p \approx 0.26$ at $z = 6$ to $p \approx 0.79$ at $z = 0$.

Coevolution of star formation and black hole accretion: Black hole accretion rates correlate with star formation rates at all epochs. At high redshift ($z \gtrsim 3$), black holes accrete efficiently with typical rates $\dot{M} \sim 10^{-3}$ to 10^{-2} M$_\odot$ yr^{-1}, and these actively accreting systems reside in star-forming galaxies. At low redshift ($z \lesssim 1$), the accretion rate distribution broadens significantly. AGN feedback decouples them by removing gas and quenching both star formation and black hole growth.

AGN-driven quenching: Massive galaxies ($M_\star \sim 10^{11}$ M$_\odot$) experience rapid quenching events at $z \lesssim 1$, characterized by simultaneous drops in gas mass, star formation rate, and black hole accretion rate occurring within ∼ 160 − 320 Myr. Lag analysis reveals no measurable temporal offset between these processes, indicating tight coupling dynamical timescale consistent with AGN feedback.

Mass-dependent feedback: Low-mass galaxies ($M_\star \sim 10^9$ M$_\odot$) show no evidence of AGN-driven quenching and remain star-forming throughout cosmic time, regulated primarly by stellar feedback. Intermediate-mass systems ($M_\star \sim 10^{10}$ M$_\odot$) exhibit episodic feedback but continue to accrete gas and maintain star formation. Only massive systems experience permanent quenching through efficient AGN feedback.

Model comparison: Comparisons of IllustrisTNG100 with the older Illustris simulation suggests that the updated TNG feedback models produces more efficient gas removal and in massive galaxies. TNG100 galaxies experience more abrupt gas depletion events compared to their Illustris counterparts.

5.2 Physical Interpretation

The Coevolution of galaxies and black holes revealed in our analysis is causally intertwined. The star-forming main sequence shows a nearly two-order-of-magntiude decline in normalization from $z = 6$ to $z = 0$ (Table 1), reflecting the gradual exhaustion of cold gas reservoirs as the universe ages, combined with increasingly efficient feedback processes. At high redshift ($z \gtrsim 4$), nearly all galaxies with $M_\star \geq 10^7$ M$_\odot$ actively form stars, producing a tight, well-defined main sequence with minimal scatter ($\sigma \lesssim 0.25$ dex). By $z \lesssim 1$, the distribution has broadened substantially ($\sigma \gtrsim 0.35$ dex), and a prominent quiescent population emerges at sSFR $\lesssim 10^{-11}$ yr^{-1}, particularly among massive systems ($M_\star \gtrsim 10^{10}$ M$_\odot$). Concurrently, the black hole population undergoes transformation. At $z = 6$, the M_{BH} distribution is sharply peaked at the simulation's seed mass ($\sim 10^6$ M$_\odot$), with only a small tail of systems that have experienced significant growth (Figure 6). Over cosmic time, this distribution broadens and shifts to higher masses, with the normalization of the M_{BH}˘$M_\star$ relation increasing systematically from $\log_{10}$(M$_{\rm BH}$/M$_\odot$) ≈ 6.6 at $z = 6$ to ≈ 7.6 at $z = 0$ for a galaxy of $M_\star = 10^{10}$ M$_\odot$ (Table 2).

Black hole accretion rates track star formation rates closely at all epochs (Figure 7), with actively accreting black holes residing almost exclusively in star-forming galaxies. This correlation arises because both processes draw from the same reservoir of cold gas in the galaxy's central region. At high redshift, when gas fractions are high and cosmic accretion rates are maximal, both star formation and black hole growth proceed efficiently. As galaxies evolve, AGN feedback becomes increasingly important, and instead of gas supply driving both SFR and $\dot{M}$, black hole-driven outflows deplete the gas supply, quenching both processes simultaneously.

5.3 The Role of AGN Feedback in Massive Galaxy Quenching

One of the most main results of this study is the AGN-driven quenching in massive galaxies. Figures 10 and 11 demonstrate that galaxies with $M_\star \sim 10^{11}\,\mathrm{M}_\odot$ experience abrupt gas depletion events at $z \lesssim 1$, accompanied by near-simultaneous drops in star formation and black hole accretion. These events are rapid transitions occurring within $\sim 160-320$ Myr (one or two simulation snapshots). The lag analysis in Figure 12 provides quantitative support for this interpretation. The absence of measurable temporal offset between gas removal, SFR decline, and $\dot{M}$ suppression indicates that all three processes are tightly coupled on timescales shorter than the simulation's temporal resolution.

Quenching is also mass-dependent. Low-mass galaxies ($M_\star \sim 10^9\,\mathrm{M}_\odot$) show no evidence of AGN-driven quenching as their gas reservoirs remain stable, and they continue forming stars throughout cosmic time. This is expected given their lower black hole masses ($M_{BH} \lesssim 10^7\,\mathrm{M}_\odot$ typically) and the weaker coupling between AGN feedback and the interstellar medium at shallow potential wells. Intermediate-mass galaxies ($M_\star \sim 10^{10}\,\mathrm{M}_\odot$) occupy a transitional regime. They exhibit episodic drops in gas content and star formation but typically recover, suggesting that AGN feedback in these systems is not yet strong enough to permanently quench star formation.

5.4 Comparison with Observations and Implications

Our simulated $M_{\rm BH} - M_\star$ relation (Figure 5, Table 2) shows good agreement with the classical Kormendy & Ho (2013) relation at low redshift ($z \lesssim 1$). The (Kormendy & Ho 2013) relation, derived from observations of local galaxies, relates black hole mass to bulge mass. At high redshift ($z \geq 2$), however, our simulated relation deviates systematically from the Kormendy & Ho (2013) fit, with $M_{\rm BH}$ values clustering near the seeding mass of $\sim 10^6\,\mathrm{M}_\odot$ (visible as the dense concentration of points at the lower left of the high-z panels in Figure 5). This deviation reflects a fundamental aspect of the IllustrisTNG simulation: the model does not simulate the physical formation of black holes, but instead employs a seeding prescription that places black holes with a fixed seed mass of $10^6\,\mathrm{M}_\odot$ at the centers of dark matter halos once they exceed a threshold mass. At high redshift, most galaxies have not yet experienced sufficient accretion or mergers to grow their black holes significantly beyond this seed mass, resulting in the observed clustering and the shallower slope of the $M_{\rm BH} - M_\star$ relation ($p \approx 0.26$ at $z = 6$ compared to $p \approx 0.79$ at $z = 0$, Table 2). As cosmic time progresses and black holes grow through gas accretion and galaxy mergers, the population gradually converges toward the observed local relation, which was one of the calibration targets for the TNG simulation physics.

The observational data from Reines & Volonteri (2015) (red points in Figure 5, $z = 0$ and $z = 0.5$ panels) provide an important comparison with our simulated relation. Their sample comprises 262 broad-line AGN in the nearby universe ($z < 0.055$) and 79 galaxies with dynamical black hole masses drawn from the Sloan Digital Sky Survey (SDSS), with black hole masses estimated using standard virial techniques. The observed galaxies exhibit a correlation between M_{BH} and $M_\star$, confirming the fundamental relationship between black hole mass and host galaxy stellar mass. However, the data show large scatter and do not fall precisely on the same sequence as our simulated best-fit relation.

This offset and increased scatter relative to the tight simulation relation likely arise from several factors. First, the Reines & Volonteri (2015) sample specifically targets AGN rather than an unbiased galaxy population. This selection may introduce biases. Furthermore, the broader scatter in observations compared to the tight simulation relation may indicate that the real universe exhibits greater diversity in black hole-galaxy evolutionary pathways than captured by TNG. Despite these differences in the detailed distribution, both our simulations and the Reines & Volonteri (2015) observations demonstrate the same underlying trend that more massive galaxies host more massive black holes, underscoring the fundamental connection between black hole growth and galaxy assembly.

The star-forming main sequence in IllustrisTNG50 shows agreement with the empirically calibrated Speagle et al. (2014) relation across the full redshift range $0 \leq z \leq 6$. The slope of the sSFR $-$ M$_\star$ relation approaches unity at $z \sim 1-2$ (Table 1), indicating that specific star formation rates are nearly independent of stellar mass during the epoch of peak cosmic star formation, in line with observations. The systematic decline in normalization from $z = 6$ to $z = 0$ also matches observational trends. However, the scatter in the SFMS increases from $\sigma \approx 0.18$ dex at $z = 6$ to $\sigma \approx 0.43$ dex at $z = 0$ in our simulations (Table 1).

5.5 Future Outlook

Our analysis provides an overall picture of the joint evolution of galaxies and their supermassive blackholes in cosmological simulations. Black holes and galaxies grow along side each other during the gas-rich early universe, coupled by their shared fueled supply. As black holes accumulate mass and AGN feedback becomes efficient, massive galaxies transition from star-forming to quiescent states through rapid gas expulsion. This feedback-regulated evolution establishes the observed scaling relations between black hole mass and galaxy properties. The IllustrisTNG simulations successfully reproduce many observed properties of galaxies and black holes, including the star-forming main sequence (SFMS), the $M_{\rm BH} - M_\star$ relation, and the emergence of quiescent galaxies.

Future work could pursue several directions. First, identifying the physical events that trigger the rapid changes in gas mass, black hole accretion rate, and star formation rate observed in Figures 10 and 11. Are these changes driven by major or minor galaxy mergers, tidal interactions and fly-bys, or internal processes such as disk instabilities? Correlating the timing of gas depletion events with merger trees and galaxies interaction histories would provide insight into the relative importance of these mechanisms. Second, our lag analysis (Figure 12) is limited by the simulation time resolution between snapshots (~ 160 Myr). Investigating cross-correlations on shorter timescales would better constrain the true response time of star formation and black hole accretion to gas removal. Third, we could investigate starburst and quiescent phases in more detail. What fraction of cosmic time do galaxies spend in each state? How do these phases correlate with merger history? Fourth, explicitly tracing individual galaxies as they transition off the star-forming main sequence would help us better study quenching. By following galaxies through the quenching transitions and examining the detailed evolution of their gas reservoirs, black hole accretion states, and feedback properties, we could better understand the connection between different feedback modes and environmental effects in driving galaxy quenching.

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